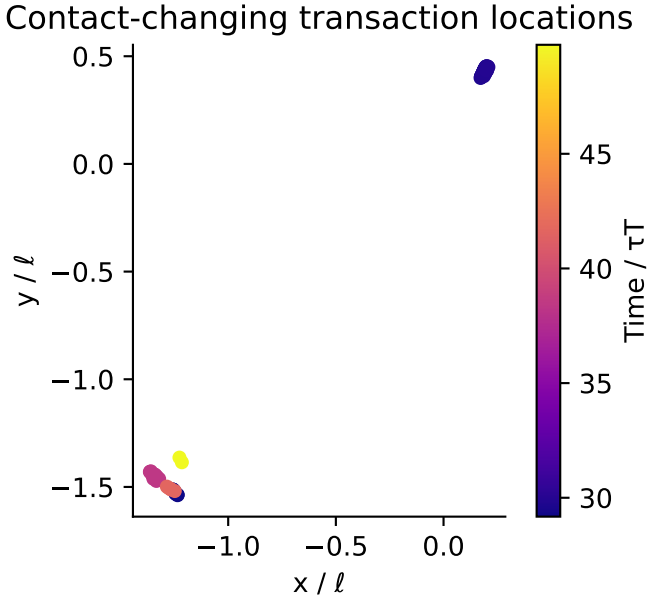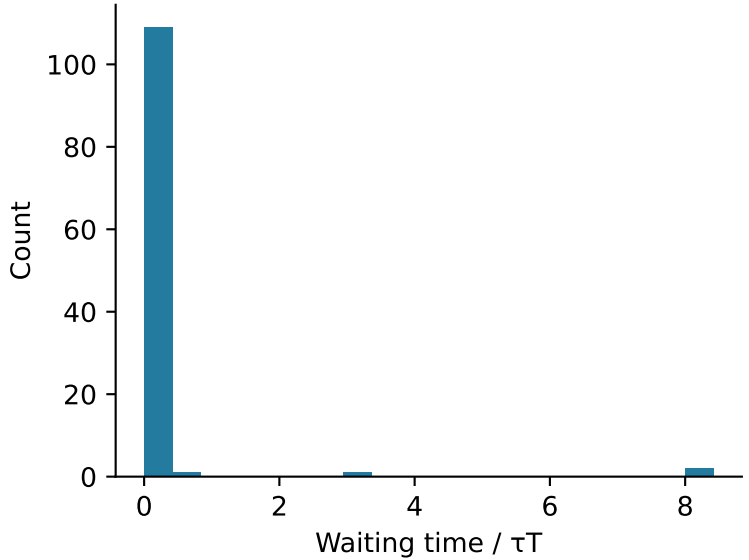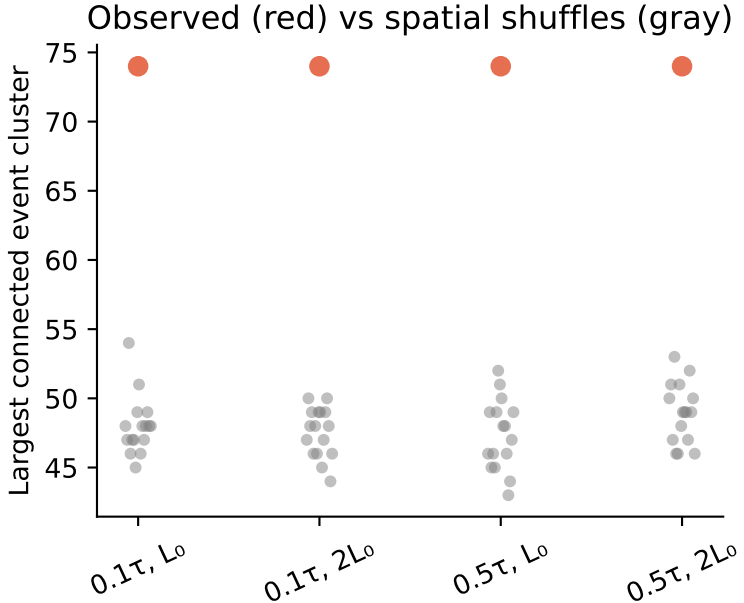Resolved neighbor changes and clustering diagnostics